

Yingjie Lei

📍 Foshan, China ✉ ylei25@jh.edu 🎓 Google Scholar 🌐 Homepage in LinkedIn 🐙 GitHub

Education

Johns Hopkins University <i>M.S.E. in Computer Science (Incoming)</i>	Starting Aug 2026
South China Normal University – Aberdeen Institute of Data Science & AI <i>B.Eng. & B.Sc. (Hons) in Artificial Intelligence</i>	Sep 2022 – Jun 2026
◦ Dual-degree program in Artificial Intelligence ◦ CN Degree: B.Eng. in Artificial Intelligence, South China Normal University (Project 211) ◦ UK Degree: B.Sc. (Hons) in Artificial Intelligence, University of Aberdeen; graduated with First Class Honours ◦ GPA: 90.9/100 (CN), 20.23/22 (UK); Rank: 5/88 (Top 5.7%) ◦ Core Courses: Machine Learning, Data Mining and Visualization, Robot Technology, Languages and Computability, Discrete Mathematics, Probability and Mathematical Statistics	

Research Interests

Representation learning, world models, and embodied AI.

Publications & Preprints

PrefBench: Evaluating Zero-Shot LLM Agents in Hidden-Preference Personalized Pricing Negotiations ↗	May 2026
Yingjie Lei	
<i>arXiv preprint, arXiv:2605.22855, 2026.</i>	
Draft-and-Target Sampling for Video Generation Policy ↗	Under Review
Qikang Zhang, Yingjie Lei , Wei Liu, Daochang Liu	
Under review, 2026.	
Test Time Training for 4D Medical Image Interpolation ↗	Apr. 2025
Qikang Zhang*, Yingjie Lei *, Zihao Zheng, Ziyang Chen, Zhonghao Xie	
IJCNN 2025, Oral Presentation.	

Research Experience

Undergraduate Thesis: Personalized Pricing of Customized Car Features <i>Supervised by Dr. Binod Bhattacharai; Thesis grade: 92/100; Outstanding Thesis (Top ~3%)</i>	SCNU / Aberdeen Jan. 2026 – Apr. 2026
◦ Formulated hidden-preference personalized pricing as a POMDP with latent buyer traits, multi-round negotiation, and profit-oriented sequential decision making. ◦ Built a reproducible simulator benchmark with public-data-grounded persona generation, explicit observable/hidden trait separation, fixed evaluation splits, and shared episode-level metrics. ◦ Implemented heuristic and learning-based seller agents, including PPO and DreamerV3, and ran ablations on reward design, semantic features, and inference-time adaptation. ◦ Refined the thesis into PrefBench ↗, a preprint for evaluating LLM agents under hidden buyer preferences.	
Draft-and-Target Sampling for Video Generation Policy <i>Second Author, manuscript under review</i>	SCNU Apr. 2025 – Sep. 2025
◦ Proposed Draft-and-Target Sampling (DTS), a training-free acceleration framework for diffusion-based video generation that uses dual denoising trajectories within a single model. ◦ Introduced Token Chunking and Progressive Acceptance to reduce redundant resampling while preserving output fidelity.	

- Achieved up to **2.1× speed-up** on *Meta-World*, *iThor*, *Libero* with minimal accuracy drop, and surpassed baselines in accuracy on *iThor*.
- Studied the method in embodied-style video-policy settings, emphasizing inference efficiency for downstream decision-making rather than only generic video generation quality.
- **Personal Contributions:** Implemented and refined the core pipeline, ran ablations and benchmarks on *Meta-World* and *iThor*, organized results, created task visualizations, and contributed to manuscript writing and revision.

Test Time Training for 4D Medical Image Interpolation

SCNU

Co-first Author, paper published in the *Proceedings of IJCNN 2025 (Oral)*

Sep. 2024 – Feb. 2025

- Proposed a **Test Time Training (TTT)** framework for 4D medical image interpolation to handle distribution shift without requiring target-domain labels.
- Designed a shared-encoder adaptation mechanism using self-supervised tasks, including *rotation prediction* and *3D-MAE reconstruction*, during inference.
- Achieved state-of-the-art results on **Cardiac MRI** and **4D-Lung CT** (**33.7dB** and **34.0dB PSNR**) with improved robustness under distribution shift.
- Framed the method as a reusable adaptation template for downstream medical vision tasks such as segmentation and registration under shifting test distributions.
- **Personal Contributions:** Implemented the core 3D-MAE module, ran experiments and metric evaluation, created visualizations, improved code clarity, and contributed to manuscript writing and revision.

Selected Course Projects

LLM-Based Intelligent Schedule Assistant (Score: 99/100, UK Grading)

Feb 2025 – Jun 2025

- Led a 6-person team to build an LLM-powered scheduling assistant; designed the system architecture, implemented the Python backend, and integrated the DeepSeek and Google Maps APIs, resulting in the highest course score of 99/100.

Self-Supervised Learning with SimCLR and MAE (Score: 90/100, UK Grading)

Feb 2025 – Apr 2025

- Implemented and compared **SimCLR** and **MAE** with a ViT backbone on Imagenette; **MAE** achieved **89.6%** Top-1 accuracy with stronger F1 and better efficiency, resulting in a project score of 90/100.

Honors & Awards

SCNU Comprehensive Scholarship (3rd Prize)

Apr 2026

Merit-based award recognizing strong academic record and overall performance

SCNU Outstanding Student Scholarship

Mar 2025

Awarded to top students with excellent academic performance (Top 8% in cohort)

SCNU-UoA Joint Institute Scholarship (3rd Prize)

Oct 2024

Granted for outstanding GPA performance, ranked 5th out of 86 students

SCNU Comprehensive Scholarship (3rd Prize)

Mar 2024

Merit-based award recognizing strong academic record and overall performance

Skills & Qualifications

- **Tests:** IELTS Academic 7.5 (L 8.5, R 8.5, W 6.5, S 7.0); GRE 319 (Q 170, V 149, AW 3.5)
- **Programming:** Python (Proficient, Research/Project Experience), Java (Academic Coursework).
- **ML/DL:** PyTorch (Solid Research Experience), HuggingFace, Diffusion Models, Transformers, CNNs, Reinforcement Learning, Test Time Training, Classic ML Algorithms (e.g., Regression, Clustering).
- **Data Science/Tools:** Jupyter Notebook, NumPy, pandas, Scikit-learn, Matplotlib, Seaborn, Data Preprocessing, Exploratory Data Analysis (EDA), Clustering, Data Visualization.
- **Productivity/Documentation:** LaTeX (Academic Writing/Reports), Git (Version Control), Zotero (Research Data and Citation Management), Microsoft 365.
- **Languages:** Mandarin (Native), Cantonese (Native), English (Fluent)